IE 206: Introduction to AI & ML

Stochastic Gradient Descent

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SGD Introduction

- Gradient descent is an iterative optimization procedure in which at each step we improve the solution by taking a step along the negative of the gradient of the function to be minimized at the current point. We typically minimise the risk function $L_D(h) = \mathbb{E}_{z\sim D}[l(h, z)]$. Let's refer to hypotheses as vectors w then we try to minimize the risk function $L_D(w)$
- We are minimizing the risk function, and since we do not know D we also do not know the gradient of $L_D(w)$.
- SGD circumvents this problem by allowing the optimization procedure to take a step in a random direction (appropriately).
- SGD is an efficient algorithm and enjoys the same sample complexity as the regularized risk minimization rule.

Gradient Descent

The gradient of a differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ at $\mathbf{w}$, denoted $\nabla f(\mathbf{w})$, is the vector of partial derivatives of f, namely,

$$\nabla f(\mathbf{w}) = \left(\frac{\partial f(\mathbf{w})}{\partial w_1}, \ldots, \frac{\partial f(\mathbf{w})}{\partial w_d} \right).$$

Gradient descent is an iterative algorithm. We start with an initial value of $\mathbf{w}$ (say, $\mathbf{w}^{(1)} = 0$). Then, at each iteration, we take a step in the direction of the negative of the gradient at the current point. That is, the update step is

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \nabla f(\mathbf{w}^{(t)}), \tag{1}$$

where $\eta > 0$ is a parameter called learning rate.

Gradient Descent

Another way to motivate gradient descent is by relying on Taylor approximation. The gradient of f at $\mathbf{w}$ yields the first-order Taylor approximation of f around $\mathbf{w}$ by

$$f(\mathbf{u}) \approx f(\mathbf{w}) + \langle \mathbf{u} - \mathbf{w}, \nabla f(\mathbf{w}) \rangle.$$

When f is convex, this approximation lower bounds f, that is,

$$f(\mathbf{u}) \geq f(\mathbf{w}) + \langle \mathbf{u} - \mathbf{w}, \nabla f(\mathbf{w}) \rangle.$$

Therefore, for $\mathbf{w}$ close to $\mathbf{w}^{(t)}$, we have that

$$f(\mathbf{w}) \approx f(\mathbf{w}^{(t)}) + \langle \mathbf{w} - \mathbf{w}^{(t)}, \nabla f(\mathbf{w}^{(t)}) \rangle.$$

Gradient Descent

Hence, we can minimize the approximation of $f(\mathbf{w})$. However, the approximation might become loose for $\mathbf{w}$ far away from $\mathbf{w}^{(t)}$. Therefore, we would like to minimize jointly the distance between $\mathbf{w}$ and $\mathbf{w}^{(t)}$ and the approximation of f around $\mathbf{w}^{(t)}$.

$$\mathbf{w}^{(t+1)} = \arg\min_{\mathbf{w}} \left[\frac{1}{2}\|\mathbf{w} - \mathbf{w}^{(t)}\|^2 + \eta \left(f(\mathbf{w}^{(t)}) + \langle \mathbf{w} - \mathbf{w}^{(t)}, \nabla f(\mathbf{w}^{(t)}) \rangle \right) \right]$$

Parameter η controls the trade-off between the two terms, we obtain the update rule
Solving the preceding equation same update rule as in Equation 1 in previous slides.

Subgradients

- The gradient descent (GD) algorithm requires that the function f be differentiable.
- We generalize the discussion beyond differentiable functions. We will show that the GD algorithm can be applied to nondifferentiable functions by using a so-called *subgradient* of $f(\mathbf{w})$ at $\mathbf{w}^{(t)}$, instead of the gradient.

Subgradients

To motivate the definition of subgradients, recall that for a convex function f, the gradient at $\mathbf{w}$ defines the slope of a tangent that lies below f, that is,

$$\forall \mathbf{u}, \quad f(\mathbf{u}) \geq f(\mathbf{w}) + \langle \mathbf{u} - \mathbf{w}, \nabla f(\mathbf{w}) \rangle. \tag{1}$$

An illustration is given on the left-hand side of Figure 14.2.
The existence of a tangent that lies below f is an important property of convex functions, which is, in fact, an alternative characterization of convexity.
Lemma: Let S be an open convex set. A function $f : S \to \mathbb{R}$ is convex if and only if for every $\mathbf{w} \in S$ there exists $\mathbf{v}$ such that

$$\forall \mathbf{u} \in S, \quad f(\mathbf{u}) \geq f(\mathbf{w}) + \langle \mathbf{u} - \mathbf{w}, \mathbf{v} \rangle. \tag{2}$$

Subgradients

Definition(Subgradients): A vector $\mathbf{v}$ that satisfies Equation 3 is called a *subgradient* of f at $\mathbf{w}$. The set of all subgradients of f at $\mathbf{w}$ is called the *differential set* and is denoted by $\partial f(\mathbf{w})$.

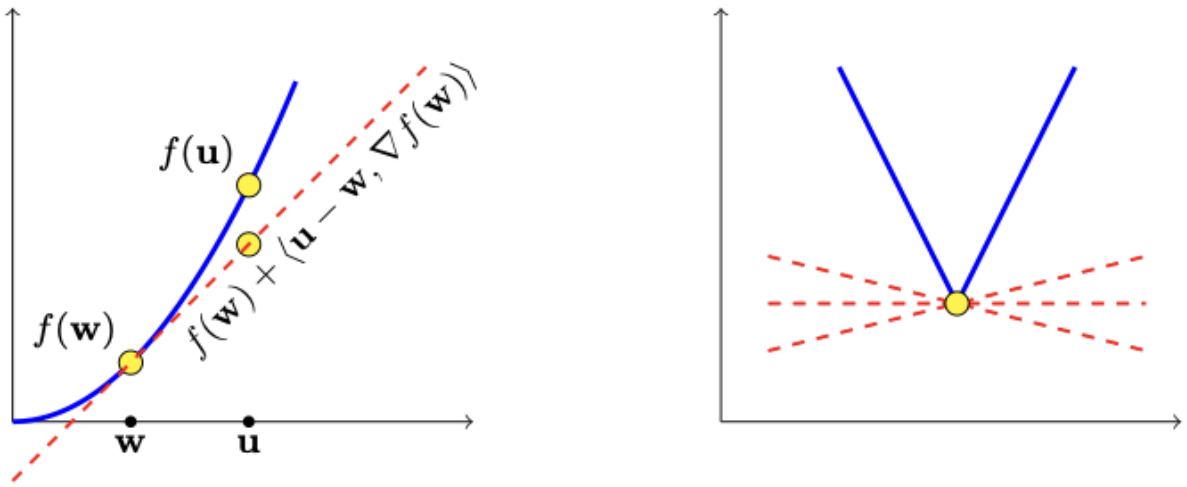

Figure: Left: The right-hand side of Equation 2 is the tangent of f at w. For a convex function, the tangent lower bounds f. Right: Illustration of several subgradients of a nondifferentiable convex function.

Calculating Subgradients

Claim 1 : If f is differentiable at $\mathbf{w}$, then $\partial f(\mathbf{w})$ contains a single element – the gradient of f at $\mathbf{w}$, $\nabla f(\mathbf{w})$.

Example (The Differential Set of the Absolute Function): Consider the absolute value function $f(x) = |x|$. Using Claim 1, we can easily construct the differential set for the differentiable parts of f, and the only point that requires special attention is $x_0 = 0$. At that point, it is easy to verify that the subdifferential is the set of all numbers between -1 and 1. Hence:

$$\partial f(x) = \begin{cases} \{1\}, & \text{if } x > 0 \\ \{-1\}, & \text{if } x < 0 \\ [-1, 1], & \text{if } x = 0 \end{cases}$$

For many practical uses, we do not need to calculate the whole set of subgradients at a given point, as one member of this set would suffice.

Subgradient of a Maximum Function

Claim 2: Let $g(\mathbf{w}) = \max_{i \in [r]} g_i(\mathbf{w})$ for r convex differentiable functions $g_1, \ldots, g_r$. Given some $\mathbf{w}$, let $j \in \arg\max_i g_i(\mathbf{w})$. Then $\nabla g_j(\mathbf{w}) \in \partial g(\mathbf{w})$.
Proof: Since g_j is convex, we have that for all $\mathbf{u}$:

$$g_j(\mathbf{u}) \geq g_j(\mathbf{w}) + \langle \mathbf{u} - \mathbf{w}, \nabla g_j(\mathbf{w}) \rangle.$$

Since $g(\mathbf{w}) = g_j(\mathbf{w})$ and $g(\mathbf{u}) \geq g_j(\mathbf{u})$, we obtain:

$$g(\mathbf{u}) \geq g(\mathbf{w}) + \langle \mathbf{u} - \mathbf{w}, \nabla g_j(\mathbf{w}) \rangle.$$

which concludes our proof. $\square$

Subgradient of Hinge Loss and Lipschitz functions

Hinge Loss: If we use the Claim 2, we get the subgradient of hinge loss , $\mathbf{v}$ as follows

$$\mathbf{v} \begin{cases} \mathbf{0} & \text{if } 1 - y\langle \mathbf{w}\ ,\mathbf{x}\rangle \leq 0 \\ -y\mathbf{x} & \text{if } 1 - y\langle \mathbf{w}\ ,\mathbf{x}\rangle > 0 \end{cases}$$

Lemma: Let A be a convex open set and let $f : A \to \mathbb{R}$ be a convex function. Then, f is ρ-Lipschitz over A iff for all $\mathbf{w} \in A$ and $\mathbf{v} \in \partial f(\mathbf{w})$ we have that:

$$\|\mathbf{v}\| \leq \rho.$$

Stochastic Gradient Descent (SGD)

In stochastic gradient descent we do not require the update direction to be based exactly on the gradient. Instead, we allow the direction to be a random vector and only require that its *expected value* at each iteration will equal the gradient direction. Or, more generally, we require that the expected value of the random vector will be a subgradient of the function at the current vector.

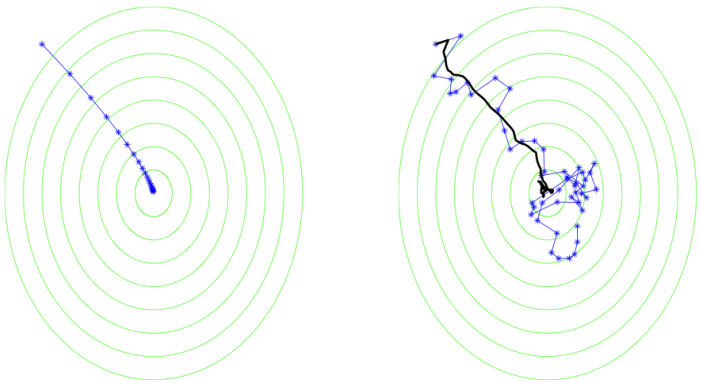

Figure: Gradient descent on the left and stochastic gradient descent on the right. In the right figure, the black line depicts the average value of $\mathbf{w}$

Algorithm for SGD

Stochastic Gradient Descent (SGD) for minimizing $f(\mathbf{w})$

parameters: Scalar $\eta > 0$, integer $T > 0$
initialize: $\mathbf{w}^{(1)} = 0$
for $t = 1, 2, \ldots, T$

- Choose $\mathbf{v}_t$ at random from a distribution such that $\mathbb{E}[\mathbf{v}_t | \mathbf{w}^{(t)}] \in \partial f(\mathbf{w}^{(t)})$.
- Update $\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \mathbf{v}_t$.

output $\bar{\mathbf{w}} = \frac{1}{T} \sum_{t=1}^{T} \mathbf{w}^{(t)}$

Variants of SGD

A. Using Variable Step Size:
Decrease the step size as a function of t. Instead of using η for update, we use η_t. For example, we can take $\eta_t = \frac{B}{\rho\sqrt{t}}$
B. Averaging Techniques:
Use sophisticated averaging schemes which can improve the convergence speed

Variants of SGD: For Strongly Convex function

Claim : If f is λ-strongly convex, then for every $\mathbf{w}, \mathbf{u}$ and $\mathbf{v} \in \partial f(\mathbf{w})$, we have

$$\langle \mathbf{w} - \mathbf{u}, \mathbf{v} \rangle \geq f(\mathbf{w}) - f(\mathbf{u}) + \frac{\lambda}{2}\|\mathbf{w} - \mathbf{u}\|^2.$$

SGD algorithm for minimizing a λ-strongly convex function
Goal: Solve $\min_{\mathbf{w} \in \mathcal{H}} f(\mathbf{w})$
Parameter: T
Initialize: $\mathbf{w}^{(1)} = \mathbf{0}$
for $t = 1, \ldots, T$ **do**

- Choose a random vector $\mathbf{v}_t$ such that $\mathbb{E}[\mathbf{v}_t | \mathbf{w}^{(t)}] \in \partial f(\mathbf{w}^{(t)})$
- Set $\eta_t = \frac{1}{\lambda t}$
- Set $\mathbf{w}^{(t+\frac{1}{2})} = \mathbf{w}^{(t)} - \eta_t \mathbf{v}_t$
- Set $\mathbf{w}^{(t+1)} = \arg\min_{\mathbf{w} \in \mathcal{H}} \|\mathbf{w} - \mathbf{w}^{(t+\frac{1}{2})}\|^2$

Output: $\bar{\mathbf{w}} = \frac{1}{T} \sum_{t=1}^{T} \mathbf{w}^{(t)}$

Adding Projection step for SGD

Let $\mathcal{H} = \mathbf{w} : \|\mathbf{w}\| \leq B$ **A. Projected Stochastic Gradient Descent (Projected SGD)**

The basic idea is to add a *projection step*, where we first subtract a subgradient from the current value of $\mathbf{w}$ and then project the resulting vector onto a feasible set $\mathcal{H}$.

1. $\mathbf{w}^{(t+\frac{1}{2})} = \mathbf{w}^{(t)} - \eta \mathbf{v}_t$
2. $\mathbf{w}^{(t+1)} = \arg\min_{\mathbf{w} \in \mathcal{H}} \|\mathbf{w} - \mathbf{w}^{(t+\frac{1}{2})}\|$

The projection step replaces the current value of $\mathbf{w}$ by the vector in $\mathcal{H}$ closest to it.

Adding Projection step for SGD

Lemma (Projection Lemma): *Let $\mathcal{H}$ be a closed convex set and let* $\mathbf{v}$ *be the projection of* $\mathbf{w}$ *onto $\mathcal{H}$, namely,*

$$\mathbf{v} = \arg\min_{\mathbf{x}\in\mathcal{H}} \|\mathbf{x} - \mathbf{w}\|^2.$$

Then for every $\mathbf{u} \in \mathcal{H}$

$$\|\mathbf{w}\text{-}\mathbf{u}\|^2 = \|\mathbf{v}\text{-}\mathbf{u}\|^2 \geq 0$$

Using the lemma, we can easily adapt the analysis of SGD to the case in which we add projection steps on a closed and convex set. Simply note that for every t,

$$\|\mathbf{w}^{(t+1)} - \mathbf{w}^\star\|^2 - \|\mathbf{w}^{(t)} - \mathbf{w}^\star\|^2$$

$$= \|\mathbf{w}^{(t+1)} - \mathbf{w}^\star\|^2 - \|\mathbf{w}^{(t+\frac{1}{2})} - \mathbf{w}^\star\|^2 + \|\mathbf{w}^{(t+\frac{1}{2})} - \mathbf{w}^\star\|^2 - \|\mathbf{w}^{(t)} - \mathbf{w}^\star\|^2$$

$$\leq \|\mathbf{w}^{(t+\frac{1}{2})} - \mathbf{w}^\star\|^2 - \|\mathbf{w}^{(t)} - \mathbf{w}^\star\|^2.$$

Convex-Lipschitz bounded problems

- Consider a convex-Lipschitz-bounded learning problem with parameters ρ, B.
- Then, for every $\epsilon > 0$, if we run the SGD method for minimizing $L_{\mathcal{D}}(\mathbf{w})$ with a number of iterations (i.e., number of examples) $T \geq \frac{B^2\rho^2}{\epsilon^2}$ and with $\eta = \sqrt{\frac{B^2}{\rho^2 T}}$, then the output of SGD satisfies

$$\mathbb{E}\big[L_{\mathcal{D}}(\bar{\mathbf{w}})\big] \leq \min_{\mathbf{w}\in\mathcal{H}} L_{\mathcal{D}}(\mathbf{w}) + \epsilon.$$

- The required sample complexity is of the same order of magnitude as the sample complexity guarantee we derived for regularised loss minimisation

Minimizing Loss function with SGD

Stochastic Gradient Descent (SGD) for minimizing $L_{\mathcal{D}}(\mathbf{w})$

parameters: Scalar $\eta > 0$, integer $T > 0$
initialize: $\mathbf{w}^{(1)} = \mathbf{0}$
for $t = 1, 2, \ldots, T$
  sample $z \sim \mathcal{D}$
  pick $\mathbf{v}_t \in \partial \ell(\mathbf{w}^{(t)}, z)$
  update $\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \mathbf{v}_t$
output $\bar{\mathbf{w}} = \frac{1}{T} \sum_{t=1}^{T} \mathbf{w}^{(t)}$

Figure: Minimizing Loss function